

Setting Up Snort to Secure Your Network

Snort is a popular, open source intrusion detection system that monitors traffic in real time to detect malware. Here's a detailed explanation of how to set it up on Ubuntu and test it by generating traffic from another system.



In today's digital world, networks often face constant risks from intrusions that may lead to data theft, service outages, or even complete system compromise if not detected in time. To address these risks, many organisations use intrusion detection systems (IDS). A widely used open source IDS tool is Snort, which monitors network traffic in real time and identifies suspicious activity based on rules and patterns.

Snort works by capturing raw network packets with its packet capture module, preprocessing them to normalise and decode protocols, and then passing this data to its detection

engine. Here, the preprocessed packets are compared against a separate set of rules, which are created based on threat intelligence from sources like CVE and Bugtraq, to identify suspicious activity. If a rule matches, Snort generates alerts/logs through its output modules.

System setup

Before installing Snort, it is important to prepare the system properly. Snort works best on a clean and updated Linux environment. We will use Ubuntu as the base operating system. Ensure that the system is up to date so that all required packages remain current.

Since we will be testing FTP detection rules later, it is also necessary to have an FTP server installed and running on the Ubuntu system. This will allow you to generate FTP traffic (USER, PASS, STOR commands, etc) for testing Snort's dynamic behaviour. The popular option is *vsftpd*, which can be installed via *apt*.

A stable internet connection is also necessary since several dependencies need to be downloaded during the installation. It is recommended to have root or sudo access to the system because most installation and configuration steps require administrative privileges. With these preparations done, the